

Structural Response to Fundamental Constant Deformations

*Operator Algebra, Phase Transitions, and Invariance Classes
in the UNNS Substrate*

Instruments: STRUC-I v1.0.4 × STRUC-PERC-I v2.4.0

Corpus: 903 constant-deformation runs · 30 systems · 6 domain families

Constants: α (fine-structure) · μ (proton-to-electron mass ratio)

Framework: Universal Structural Law · Percolative Realizability Principle

Status: Empirically grounded — open conjectures identified

Date: 2026

Prior work in the UNNS Substrate Research Programme established three static structural results: the Universal Structural Law (USL) defines the admissibility domain $\mathcal{M}_{\text{adm}}$; the Percolative Realizability Principle (PRP) defines the realizability structure within $\mathcal{M}_{\text{adm}}$; and the Dual Observability Theorem establishes that admissibility ($\bar{\rho}$) and realizability ($\mathcal{C}, \kappa_{\text{conn}}$) are independent structural coordinates. These results characterise structure at fixed physical parameter values.

The present paper introduces a dynamical extension and studies how the structural state $\mathcal{S}(S) = (\bar{\rho}(S), \mathcal{R}(S))$ evolves under controlled deformation of fundamental constants α and μ across the scaling domain $\gamma \in [0.80, 1.20]$. From a corpus of 903 constant-deformation runs across 30 physical systems, we identify consistent structural response patterns and formulate eight corpus-supported results.

The central empirical finding is *operator separation*: within the tested corpus, the μ -operator is realizability-inert (127 μ -deformation runs, zero HARD verdicts, full class invariance across all tested domains), while the α -operator produces domain-specific realizability transitions. These observations support a structured decomposition of operator action into admissibility and realizability components and motivate a classification of systems into realizability response classes.

Additional corpus-supported results establish: (i) the existence of realizability phase-like transitions in selected systems under α -deformation; (ii) the internal metric depth of the FULL realizability class (a factor of 5×10^5 spread in κ_{conn} within a single class); (iii) representation-dependence of the realizability class (the same physical system returns different classes under different ladder encodings); and (iv) a corpus-level empirical exclusion of the joint state ($\bar{\rho} \rightarrow 1, \mathcal{C} = \text{HARD}$) at physical parameter values.

Together these results support the formulation of a *dynamical theory of structural response*, in which physical systems trace operator-dependent trajectories through the structural state space — characterised not only by their position at fixed parameters but by their class-governed evolution under deformation.

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1 Introduction

1.1 Prior Framework

The UNNS Substrate Research Programme has established, through systematic falsification-first empirical work across ten physical domain families and two biological systems, a dual-layer structural theory of physical systems.

The *first layer*, the Universal Structural Law (USL) [1], asserts that every persistent ordered physical sequence satisfies the admissibility inequality

$$\text{inv}(P_\varepsilon; L) \leq \nu(V_\varepsilon(L)), \quad (1)$$

where $\text{inv}(P_\varepsilon; L)$ counts inversions in the ε -persistence set and $\nu(V_\varepsilon(L))$ is the admissible inversion capacity. Empirical evaluation via STRUC-I v1.0.4 across more than 1,500 ladders produces zero hard violations at physical parameter values.

The *second layer*, the Percolative Realizability Principle (PRP) [2], defines the internal connectivity structure of admissible configurations through the vulnerability graph $G_\kappa(L)$ and the four-tier realizability taxonomy (FULL, GIANT, TAIL, HARD). The Dual Observability Theorem [3] establishes that the admissibility coordinate $\bar{\rho}$ and the realizability coordinate $\mathcal{R} = (\mathcal{C}, \kappa_{\text{conn}})$ are independent: neither determines the other.

1.2 The Gap: Static Characterisation

All prior results characterise structure at *fixed physical parameter values*. They answer: given the observed values of α , μ , α_s , α_G , what is the structural state of a system? They do not answer: how does the structural state *respond* when parameters are varied?

This distinction matters for two reasons. First, constant-deformation studies are the primary empirical tool of the UNNS programme: the four-column Alignment Matrix [1] establishes which constants are structurally active by sweeping $\gamma = c/c_{\text{phys}} \in [0.80, 1.20]$ and observing the admissibility response $\bar{\rho}(\gamma)$. Second, the realizability coordinate has not yet been studied under deformation. Whether $\mathcal{C}(S(\gamma))$ varies with γ , and whether it varies in the same way as $\bar{\rho}(\gamma)$, are open questions.

1.3 This Paper

We introduce a dynamical extension of the dual-layer theory. For each constant $c \in \{\alpha, \mu\}$, we define a *structural deformation operator* $\hat{O}_c$ acting on the structural state space and characterise its action on both structural coordinates.

The empirical basis is a corpus of 903 constant-deformation runs from STRUC-PERC-I v2.4.0: 776 α -deformation runs across 26 system/domain combinations, and 127 μ -deformation runs across 8 molecular and nuclear systems. Each run evaluates the realizability class $\mathcal{C}(S(\gamma))$ and connectivity threshold $\kappa_{\text{conn}}(S(\gamma))$ at a specific scaling γ .

The principal results are:

1. **Operator Separation** (Theorem 3.1): $\hat{O}_\mu$ is realizability-inert; $\hat{O}_\alpha$ generates domain-specific transitions.
2. **Operator Non-Commutativity** (Conjecture 3.4): $\hat{O}_\alpha \circ \hat{O}_\mu \neq \hat{O}_\mu \circ \hat{O}_\alpha$ in general.

3. **Invariance Classes** (Theorem 4.2): physical systems partition into realizability-stable, transitional, and permanently-fragmenting classes under each operator.
4. **Phase Transitions** (Theorem 5.2): realizability class changes occur at specific γ thresholds.
5. **Metric Depth of FULL** (Theorem 6.1): the FULL class has a factor of 5×10^5 internal spread in κ_{conn} .
6. **Representation Non-Invariance** (Theorem 7.1): realizability class depends on ladder encoding.
7. **Structural Exclusion Principle** (Theorem 8.1): the joint state ($\bar{\rho} \rightarrow 1, \mathcal{C} = \text{HARD}$) is empirically excluded at physical constant values.
8. **TAIL Admissibility Under Deformation** (Theorem 9.1): TAIL-class non-percolating systems can remain admissible across deformation.

1.4 Organisation

Section 2 defines the deformation framework and operator algebra. Section 3 proves operator separation. Section 4 classifies invariance regimes. Section 5 establishes phase transitions. Section 6 develops the metric structure of the FULL class. Section 7 treats representation dependence. Section 8 proves the structural exclusion principle. Section 9 addresses TAIL admissibility. Section 10 synthesises the dynamical theory. Section 11 discusses implications and open problems.

2 Setup and Notation

We adopt the notation of the companion manuscripts [2, 3].

2.1 Structural State

Definition 2.1 (Structural state). The *structural state* of an admissible configuration $S \in \mathcal{M}_{\text{adm}}$ is the pair

$$\mathcal{S}(S) = (\bar{\rho}(S), \mathcal{R}(S)),$$

where $\bar{\rho}(S) = |\mathcal{K}|^{-1} \sum_{\kappa \in \mathcal{K}} A_{\kappa}(S)$ is the mean admissibility score (structural pressure) and $\mathcal{R}(S) = (\mathcal{C}(S), \kappa_{\text{conn}}(S))$ is the realizability coordinate with class $\mathcal{C}(S) \in \{\text{FULL}, \text{GIANT}, \text{TAIL}, \text{HARD}\}$ and connectivity threshold $\kappa_{\text{conn}}(S) \in (0, \infty]$ (see [3], Definition 2.5).

Remark 2.2. By the Dual Observability Theorem [3], $\bar{\rho}(S)$ and $\mathcal{R}(S)$ are empirically independent: neither coordinate determines the other. The present paper studies how each coordinate evolves under constant deformation.

2.2 Deformation Parameter Space

Definition 2.3 (Deformation parameter and scaling). Let $c \in \{\alpha, \mu\}$ be a fundamental constant with physical value c_{phys} . The *scaling parameter* $\gamma = c/c_{\text{phys}} \in \Gamma$ sweeps the interval $\Gamma = [0.80, 1.20]$ over a discrete grid $\Gamma_d \subset \Gamma$ with $|\Gamma_d| \in [17, 41]$ depending on the

system. The *deformed system* at scaling γ is denoted $S(\gamma)$, with $S(1) = S$ the physical system.

Definition 2.4 (Admissibility and realizability response functions). For a system S and constant c , define:

$$\mathbf{A}_c(S, \gamma) = \bar{\rho}(S(\gamma)), \quad (2)$$

$$\mathbf{R}_c(S, \gamma) = \mathcal{R}(S(\gamma)) = (\mathcal{C}(S(\gamma)), \kappa_{\text{conn}}(S(\gamma))). \quad (3)$$

These are the *admissibility response function* and *realizability response function* of S under constant c .

2.3 Structural Deformation Operators

Definition 2.5 (Structural deformation operators). For each constant $c \in \{\alpha, \mu\}$, the *structural deformation operator* $\hat{O}_c$ acts on structural states:

$$\hat{O}_c(\gamma) : \mathcal{S}(S) \mapsto \mathcal{S}(S(\gamma)).$$

Specifically:

$$\hat{O}_\alpha(\gamma) : (\bar{\rho}(S), \mathcal{R}(S)) \mapsto (\bar{\rho}(S(\gamma)), \mathcal{R}(S(\gamma))), \quad \gamma = \alpha/\alpha_{\text{phys}}. \quad (4)$$

$$\hat{O}_\mu(\gamma) : (\bar{\rho}(S), \mathcal{R}(S)) \mapsto (\bar{\rho}(S(\gamma)), \mathcal{R}(S(\gamma))), \quad \gamma = \mu/\mu_{\text{phys}}. \quad (5)$$

Each operator is parameterised by a scaling $\gamma \in \Gamma$ and acts on the full two-component structural state.

Remark 2.6 (Operator projections). Each operator $\hat{O}_c(\gamma)$ has two independent projections:

$$\Pi_{\bar{\rho}}[\hat{O}_c(\gamma)] = \mathbf{A}_c(S, \gamma), \quad (6)$$

$$\Pi_{\mathcal{R}}[\hat{O}_c(\gamma)] = \mathbf{R}_c(S, \gamma). \quad (7)$$

Prior work [1] characterises the $\Pi_{\bar{\rho}}$ projection for α and μ . The present paper characterises the $\Pi_{\mathcal{R}}$ projection for both operators.

2.4 Realizability Response Modes

Definition 2.7 (Realizability response modes). Under a deformation operator $\hat{O}_c$, a system S is classified by its realizability response mode:

- *Invariant*: $\mathcal{C}(S(\gamma)) = \mathcal{C}(S)$ for all $\gamma \in \Gamma_d$.
- *Transitional*: $\mathcal{C}(S(\gamma)) \neq \mathcal{C}(S)$ for some $\gamma \in \Gamma_d \setminus \{1\}$, but $\mathcal{C}(S(1)) = \mathcal{C}(S)$.
- *Critical*: $\mathcal{C}(S(1)) \neq \mathcal{C}(S_{\text{ref}})$, where S_{ref} is a reference class (typically FULL).

3 Operator Separation

Theorem 3.1 (Operator Separation — corpus-level result). *Within the tested corpus, the operators $\hat{O}_\alpha$ and $\hat{O}_\mu$ act on distinguishable structural components:*

1. (Corpus-level μ -inertness) *Within the tested μ -deformation corpus (127 runs, 8 system/domain combinations), the μ -operator is realizability-inert: for every tested system S and every $\gamma \in \Gamma_d$,*

$$\Pi_{\mathcal{R}}[\hat{O}_{\mu}(\gamma)](S) = \mathcal{R}(S),$$

i.e. $\mathcal{C}(S(\gamma)) = \mathcal{C}(S)$ and $\kappa_{\text{conn}}(S(\gamma)) = \kappa_{\text{conn}}(S)$ for all tested γ .

2. (α -activity) *The α -operator generates non-trivial realizability evolution in at least two tested domains (atomic and geoid), establishing that its realizability projection is not the identity.*

Thus, on the present corpus, the two operators produce distinguishable realizability responses.

Empirical proof. Part 1 (μ -inertness). The μ -column corpus comprises 127 runs across 8 system/domain combinations: CO, HCl, HD, H₂, N₂ (molecular); ⁴⁸Ca, ¹⁵⁰Nd, ²⁰⁸Pb (nuclear). The scaling grid is $\gamma \in [0.80, 1.20]$ with $|\Gamma_d| = 17$ per system.

Result: **zero HARD verdicts** across all 127 runs. Every system maintains FULL percolation at every tested μ -scaling. The realizability class $\mathcal{C}$ and connectivity threshold κ_{conn} are identical at $\gamma = 1.00$ (physical) and at all tested non-physical values. The $\Pi_{\mathcal{R}}$ projection of $\hat{O}_{\mu}$ is identically the identity on the tested corpus.

Part 2 (α -activity). The α -column corpus comprises 776 runs. In the atomic domain, hydrogen fine-structure levels are HARD at all 41 tested γ -values; helium exhibits HARD verdicts in 34 of 52 runs across two ladder representations at all tested γ . In the geoid domain, Earth and Moon harmonics exhibit HARD verdicts at $\gamma = 1.10$ and $\gamma = 1.20$. In the nuclear domain, ²⁸Si exhibits HARD verdicts at $\gamma \leq 0.90$ and at $\gamma = 0.97$.

Thus $\Pi_{\mathcal{R}}[\hat{O}_{\alpha}]$ is non-trivial in at least two domains, while $\Pi_{\mathcal{R}}[\hat{O}_{\mu}]$ is trivial throughout the tested corpus. Since the two projections differ, the operators act on independent structural dimensions. $\square$

Corollary 3.2 (Corpus-level μ -inertness). *On the tested corpus, the realizability projection of the μ -operator is empirically indistinguishable from the identity:*

$$\Pi_{\mathcal{R}}[\hat{O}_{\mu}(\gamma)](S) = \mathcal{R}(S) \quad \forall (S, \gamma) \text{ in the tested corpus.}$$

This is consistent with — but does not yet establish as universal — the hypothesis that μ acts primarily on the admissibility coordinate. The tested corpus covers molecular and three nuclear systems; extension to other domains (atomic, CMB, geoid) under μ -deformation remains an open empirical question.

Corollary 3.3 (Operator inequivalence). $\Pi_{\mathcal{R}}[\hat{O}_{\alpha}] \neq \Pi_{\mathcal{R}}[\hat{O}_{\mu}]$ on the tested corpus. *The two operators are not equivalent as structural transformations.*

Conjecture 3.4 (Operator non-commutativity). Under joint deformation, the operators $\hat{O}_{\alpha}$ and $\hat{O}_{\mu}$ may fail to commute:

$$\hat{O}_{\alpha}(\gamma_{\alpha}) \circ \hat{O}_{\mu}(\gamma_{\mu}) \neq \hat{O}_{\mu}(\gamma_{\mu}) \circ \hat{O}_{\alpha}(\gamma_{\alpha}).$$

Remark 3.5 (Status of Conjecture 3.4). This conjecture is motivated by the observed inequivalence of the two operators' realizability projections in their respective marginal sweeps. However, the current corpus contains no joint (α, μ) deformation runs; the operators have only been tested independently. What is established is *operator inequivalence*: $\Pi_{\mathcal{R}}[\hat{O}_{\alpha}] \neq \Pi_{\mathcal{R}}[\hat{O}_{\mu}]$ on the corpus. This is consistent with non-commutativity but does not imply it — equal marginal projections can arise from commutative or non-commutative compositions. The conjecture therefore remains entirely open and untested. Joint (α, μ) sweeps are required to test it directly.

4 Invariance Classes

Definition 4.1 (Realizability response classes). Under a deformation operator $\hat{O}_c$, systems are assigned to one of three *realizability response classes* based on their behaviour across Γ_d :

- *Invariant* ($\mathcal{I}_c$): the realizability class is unchanged at every tested γ :

$$\mathcal{I}_c = \{S : \mathcal{C}(S(\gamma)) = \mathcal{C}(S) \forall \gamma \in \Gamma_d\}.$$

- *Transitional* ($\mathcal{T}_c$): the class changes at some non-physical γ but is the same as the reference class at $\gamma = 1$:

$$\mathcal{T}_c = \{S \notin \mathcal{I}_c : \mathcal{C}(S(1)) = \mathcal{C}(S)\}.$$

- *Fragmenting* ($\mathcal{F}_c$): the class at the physical value is not FULL (the nominal non-fragmenting class):

$$\mathcal{F}_c = \{S : \mathcal{C}(S(1)) \neq \text{FULL}\}.$$

This is a *response taxonomy*, not a strict equivalence partition: the three classes describe distinct behavioural regimes under deformation. Note that $\mathcal{I}_c$ and $\mathcal{F}_c$ can overlap (a system can be HARD-invariant, i.e. in both $\mathcal{I}_c$ and $\mathcal{F}_c$, as demonstrated by H fine-structure levels).

Theorem 4.2 (Existence and population of invariance classes). *All three realizability invariance classes are non-empty in the observed corpus:*

1. (Strong invariants, $\mathcal{I}_{\alpha}$) *The set $\mathcal{I}_{\alpha}$ is non-empty and contains systems invariant across the full α -sweep. Specifically: all 14 nuclear isotopes tested, Li, Na, CO, N₂, HCl, HD, H₂, CMB TT/TE/EE, and DESI are in $\mathcal{I}_{\alpha}$ with $\mathcal{C} = \text{FULL}$. Hydrogen fine-structure levels are in $\mathcal{I}_{\alpha}$ with $\mathcal{C} = \text{HARD}$ (a HARD-invariant system).*
2. (Transitional, $\mathcal{T}_{\alpha}$) *The set $\mathcal{T}_{\alpha}$ is non-empty. ²⁸Si is in $\mathcal{T}_{\alpha}$: HARD at $\gamma \leq 0.90$ and $\gamma = 0.97$, FULL at $\gamma = 1.00$. ²³⁸U is in $\mathcal{T}_{\alpha}$: HARD at $\gamma = 1.02$, FULL at $\gamma = 1.00$.*
3. (Fragmenting, $\mathcal{F}_{\alpha}$) *Hydrogen fine-structure levels are HARD at $\gamma = 1.00$, placing them in $\mathcal{F}_{\alpha}$.*

Under $\hat{O}_{\mu}$: all tested systems are in $\mathcal{I}_{\mu}$ with $\mathcal{C} = \text{FULL}$. The sets $\mathcal{T}_{\mu}$ and $\mathcal{F}_{\mu}$ are empty in the tested corpus.

Empirical proof. The claim follows directly from the corpus: (1) All nuclear and molecular systems, CMB, and DESI systems are FULL at all tested α -values; the class does not change, placing them in $\mathcal{I}_\alpha$. Hydrogen fs levels are HARD at all 41 tested γ -values ($\gamma \in [0.80, 1.20]$, $GR = 0.842$ constant throughout), placing them in $\mathcal{I}_\alpha$ with $\mathcal{C} = \text{HARD}$. (2) ^{28}Si is FULL at $\gamma = 1.00$ ($GR = 1.000$, $\kappa_{\text{conn}} = 5299$) and HARD at $\gamma \in \{0.80, 0.85, 0.90, 0.97\}$ ($GR \in [0.953, 0.960]$). ^{238}U is FULL at $\gamma = 1.00$ ($\kappa_{\text{conn}} = 1490$) and HARD at $\gamma = 1.02$. Both are in $\mathcal{T}_\alpha$. (3) Hydrogen fs levels at $\gamma = 1.00$ are HARD, placing them in $\mathcal{F}_\alpha$. (4) The μ -corpus contains no HARD verdicts, so $\mathcal{T}_\mu = \mathcal{F}_\mu = \emptyset$ in the tested corpus. $\square$

Remark 4.3 (HARD-invariant systems). Hydrogen fine-structure levels constitute a novel structural type: a *HARD-invariant* system, one that is permanently HARD across the entire deformation domain. This is not a deformation-induced instability but a permanent structural property of the gap distribution. The H fine-structure level set organises into principal quantum number shells ($n = 1, 2, 3, \dots$) with inter-shell gaps many orders of magnitude larger than within-shell gaps; the maximum-to-median gap ratio is $\sim 1.16 \times 10^9$. This structural feature is invariant under α -scaling: changing the fine-structure constant by $\pm 20\%$ does not alter the shell organisation that causes permanent fragmentation.

Proposition 4.4 (Invariance class prediction). *The realizability invariance class of a system under $\hat{O}_\alpha$ is determined by the structural organisation of its gap distribution, not by its admissibility pressure $\bar{\rho}$. Specifically:*

- *Systems with gap distributions dominated by a single structural hierarchy (principal quantum number shells in hydrogen fs levels) tend toward HARD-invariance.*
- *Systems with approximately uniform gap distributions (nuclear spectral ladders, molecular rovibrational ladders) tend toward FULL-invariance.*
- *Systems with intermediate structure (geoid harmonics under large α amplification, ^{28}Si at low α) can be transitional.*

5 Phase Transitions in Realizability

Definition 5.1 (Realizability class change). A *realizability class change* under operator $\hat{O}_\alpha$ occurs at $\gamma^* \in \Gamma_d$ if

$$\mathcal{C}(S(\gamma^*)) \neq \mathcal{C}(S(\gamma^* - \delta))$$

for some $\delta > 0$ such that $\gamma^* - \delta \in \Gamma_d$. When a sequence of such changes at nearby γ -values produces a persistent transition between two class regions, we call it a *phase-like transition*; isolated single-point class changes are called *instability points*. The precise sharpness of a transition is undetermined from the present discrete grid.

Theorem 5.2 (Evidence for realizability class changes under α -deformation). *The corpus contains multiple instances of realizability class changes under α -deformation, indicating the presence of phase-like structure in parameter space. Three distinct patterns are observed:*

1. (^{28}Si — multi-point transition region) *^{28}Si is FULL at $\gamma \in \{0.95, 0.96, 0.98, 0.99, 1.00, \dots, 1.20\}$ and HARD at $\gamma \in \{0.80, 0.85, 0.90\}$, with $GR \in [0.953, 0.960]$ in the HARD region. A class-change boundary lies in $\gamma \in (0.90, 0.95)$. An isolated HARD verdict also occurs*

at $\gamma = 0.97$, consistent with a narrow instability point. The precise nature of these class changes (sharp first-order transition vs. continuous crossover) is unresolved at the current grid resolution.

2. (^{238}U — single instability point) ^{238}U is *FULL* at 32 of 33 tested γ -values and *HARD* at $\gamma = 1.02$ only. This single-point *HARD* verdict constitutes an isolated instability rather than a robust transition region; dense sampling around $\gamma = 1.02$ is needed to confirm its robustness.
3. (Geoid Earth and Moon — *HARD* regime beyond physical range) *Earth and Moon geoid harmonics return HARD at $\gamma = 1.10$ and $\gamma = 1.20$ ($GR \approx 0.75\text{--}0.82$). Since $\gamma = 1.00$ is not included in the geoid corpus, these results establish the existence of a *HARD* regime at amplified α but do not, by themselves, determine whether a class change occurs at or below the physical value.*

Empirical proof. All instances are directly observed in the corpus: ^{28}Si *HARD* at $\gamma \in \{0.80, 0.85, 0.90, 0.97\}$, *FULL* otherwise; ^{238}U *HARD* at $\gamma = 1.02$, *FULL* at all other 32 tested values; Geoid Earth *HARD* at $\gamma \in \{1.10, 1.20\}$ ($GR = 0.756, 0.751$); Geoid Moon *HARD* at $\gamma \in \{1.10, 1.20\}$ ($GR = 0.824, 0.757$). In each case the *HARD* verdict is the directly observed output of STRUC-PERC-I v2.4.0; the structural interpretation (transition vs. instability point) is a secondary classification based on the observed pattern. $\square$

Corollary 5.3 (Phase-like structure in α -parameter space). *The α -parameter space Γ is not structurally uniform: realizability class varies with γ in system-specific ways. For ^{28}Si , the corpus supports the existence of at least two distinct class regions separated by a boundary in $(0.90, 0.95)$. For ^{238}U , the corpus identifies a localised instability at $\gamma = 1.02$. In both cases the physical value $\gamma = 1.00$ lies in the *FULL* region.*

Remark 5.4 (He — representation-induced apparent transition). Helium exhibits two ladder representations that return *HARD* at all tested γ -values and one that returns *FULL* at all tested values. This is not a phase transition in the sense of Definition 5.1 — the class does not change within any single representation as γ varies. Rather, it is a *representation-induced split*, treated separately in Section 7.

6 Metric Structure of the FULL Class

Theorem 6.1 (Internal metric depth of *FULL* realizability). *The *FULL* realizability class has non-trivial internal structure measurable by κ_{conn} . Across the corpus of *FULL* systems at the physical value $\gamma = 1.00$:*

$$\kappa_{\text{conn}} \in [2, 1,000,000], \quad \text{a factor of } 5 \times 10^5.$$

This spread is domain-characteristic and irreducible to the admissibility coordinate.

Empirical proof. The observed κ_{conn} values at $\gamma = 1.00$ for *FULL* systems are:

Domain	System	Constant	κ_{conn}
Molecular	HCl, N ₂	μ	2
Molecular	H ₂	μ	1,422
Nuclear	¹⁵⁰ Nd	α/μ	301
Nuclear	²⁴ Mg	α	2,771
Nuclear	⁴⁸ Ca	α/μ	13,240
Nuclear	²⁰⁸ Pb	α/μ	38,013
Atomic	Na	α	2,446
Atomic	Li	α	82,359
Atomic	He	α	1,000,000
Molecular	CO	μ	714,804
CMB	TT	α	134,683
CMB	EE	α	859,341

The minimum is HCl and N₂ at $\kappa_{\text{conn}} = 2$; the maximum is He at $\kappa_{\text{conn}} = 10^6$ (adaptive extension limit) and CMB EE at $\kappa_{\text{conn}} = 859,341$. The ratio $10^6/2 = 5 \times 10^5$. All these systems are classified FULL (GR = 1.000) and are admissible (zero USL violations). The factor- 5×10^5 spread occurs within a single realizability class at a fixed admissibility verdict. $\square$

Corollary 6.2 (κ_{conn} as a realizability metric). *The FULL class is not homogeneous: it carries a metric structure given by κ_{conn} . Two FULL systems can differ by five orders of magnitude in their connectivity depth. This metric is not captured by the realizability class alone ($\mathcal{C} = \text{FULL}$ for all), nor by the admissibility coordinate ($\bar{\rho}$ values range from 0.10 to 0.96 within the FULL class). The triple $(\bar{\rho}, \mathcal{C}, \kappa_{\text{conn}})$ provides strictly more structural information than either $\bar{\rho}$ alone or $(\bar{\rho}, \mathcal{C})$ alone.*

Proposition 6.3 (Domain fingerprints in κ_{conn}). *The κ_{conn} values of FULL systems at physical constants cluster by domain family:*

- Molecular diatomics (μ -column): $\kappa_{\text{conn}} \in \{2, 1422, 10203, 714804\}$ — spanning six orders of magnitude within the molecular domain alone. HCl and N₂ connect immediately at $\kappa_{\text{conn}} = 2$; CO requires extreme delay at $\kappa_{\text{conn}} \sim 7 \times 10^5$.
- Nuclear (α -column): $\kappa_{\text{conn}} \in [300, 38013]$ — two orders of magnitude, with lighter isotopes generally connecting at lower κ_{conn} than heavier deformed nuclei.
- CMB spectra (α -column): $\kappa_{\text{conn}} \in [65708, 859341]$ — the ordering $TT < TE < EE$ reflects progressively sparser gap distributions in the polarization spectra.

7 Representation Dependence

Theorem 7.1 (Realizability non-invariance under ladder encoding). *The realizability class $\mathcal{C}(S)$ of a physical system S is not invariant under the choice of ladder encoding. Different admissible ladder constructions of the same physical system can yield different realizability classes.*

Constructive proof. Helium provides a direct counterexample. Three ladder constructions of the same physical system (the helium atom under α -deformation) are evaluated:

- *Construction 1* (383-gap ladder, 54 outliers): FULL percolation at every tested $\gamma \in [0.80, 1.20]$, $\kappa_{\text{conn}} = 10^6$ (adaptive extension limit).
- *Construction 2* (841-gap ladder, 169 outliers): HARD fragmentation at every tested γ , $\text{GR} \in [0.79, 0.86]$.
- *Construction 3* (842-gap ladder, 174 outliers): HARD fragmentation at every tested γ , $\text{GR} \in [0.82, 0.85]$.

All three constructions are admissible (zero USL violations, all $\text{GR} < 1$ for HARD constructions is a realizability property, not an admissibility violation). The admissibility coordinate $\bar{\rho}$ does not distinguish them; the realizability coordinate does, categorically. $\square$

Corollary 7.2 (Ladder construction as structural variable). *Ladder construction is not a mere computational detail but a structural variable of the realizability theory. Different encodings of the same physical system access different structural facets of its gap distribution. The finer the encoding (more gaps), the more internal sub-structure is exposed — and the more likely it is that extreme outlier gaps (inter-shell transitions in helium) will produce HARD fragmentation.*

Proposition 7.3 (Canonical ladder problem). *The helium results indicate that realizability class depends on the choice of ladder encoding for the same physical system. This motivates the formulation of a canonical ladder construction that would yield a representation-stable realizability class. A natural candidate would control outlier gap counts while retaining all physically distinct transitions, but such a construction faces several unresolved difficulties: (i) “physically distinct” is not formally defined in the present framework; (ii) minimising outlier counts may not be unique; (iii) aggressive outlier suppression could discard genuine structural content.*

Formulating a canonical ladder construction and determining whether realizability class becomes representation-invariant under it is an open problem. We therefore do not offer a definition here but note the problem as a primary target for future theoretical development (Section 11, Problem 3).

8 The Structural Exclusion Principle

Theorem 8.1 (Structural exclusion — corpus statement). *Within the present corpus of 903 runs, no physical system simultaneously exhibits high admissibility pressure ($\bar{\rho} \geq 0.8$) and HARD realizability at its physical constant values ($\gamma = 1.00$).*

More precisely: all HARD verdicts at $\gamma = 1.00$ in the corpus arise from H fine-structure levels (admissibility class not boundary-adjacent: $\bar{\rho}$ is not in the CRITICAL or BOUNDARY zone) or He representations with large outlier counts (also not boundary-adjacent). Every boundary-adjacent system tested ($\bar{\rho} \geq 0.8$, including Zeeman ladders at $\bar{\rho} = 0.9585$) returns FULL at $\gamma = 1.00$.

Empirical proof. The corpus contains three sources of HARD verdicts at $\gamma = 1.00$:

1. H fine-structure levels: HARD at all tested γ . The admissibility coordinate places this system outside the boundary-adjacent zone.
2. He constructions 2 and 3: HARD at all tested γ . Also not boundary-adjacent.
3. All boundary-adjacent systems with $\bar{\rho} \geq 0.8$: FULL at $\gamma = 1.00$ without exception.

No system with both $\bar{\rho} \geq 0.8$ and $\mathcal{C} = \text{HARD}$ appears at $\gamma = 1.00$ anywhere in the 903-run corpus. This corpus-level exclusion is consistent with — and anticipated by — the Forbidden State Theorem of [3]: high structural pressure requires a dense bulk of near-equal gaps, which is precisely the condition favouring broad connectivity in the vulnerability graph. Whether the exclusion holds beyond the tested corpus is an open empirical question. $\square$

Remark 8.2 (Exclusion under deformation). The corpus-level exclusion extends to non-physical γ -values: HARD verdicts at non-physical γ also occur only in systems with low-to-moderate admissibility pressure. This is consistent with the exclusion being a structural property of the joint (admissibility, realizability) state space, but the evidence is corpus-relative throughout.

9 TAIL Admissibility Under Deformation

Theorem 9.1 (TAIL class admissibility under constant deformation). *TAIL-class non-percolating systems can be admissible under constant deformation. Two corpus instances are observed:*

1. *CMB TT (α -deformation): one TAIL verdict at a non-physical γ -value; $A_{\kappa}^{\min} = 1.000$ (no USL violation).*
2. *CMB TE (α -deformation): one TAIL verdict at a non-physical γ -value; $A_{\kappa}^{\min} = 1.000$.*

In both cases, TAIL non-percolation does not imply, and is not associated with, any admissibility violation.

Empirical proof. CMB TT returns 36 FULL verdicts and 1 TAIL verdict across 37 tested α -values. CMB TE returns 30 FULL verdicts and 1 TAIL verdict across 31 tested values. In each case, the TAIL verdict occurs at a non-physical γ where $A_{\kappa}^{\min} = 1.000$ (exact ceiling). Theorem 1 of the revised PRP [2] applies only to HARD-class fragmentation; TAIL does not trigger it. The TAIL verdict is therefore admissible, consistent with the general principle that non-percolation is not equivalent to violation. $\square$

Corollary 9.2 (Deformation-induced TAIL is a realizability property). *When a system transitions from FULL to TAIL under constant deformation, this is a change in its realizability coordinate — not in its admissibility coordinate. The system remains admissible throughout. TAIL emergence under deformation signals the onset of connectivity delay (an outlier gap growing beyond the threshold), not structural failure.*

10 Synthesis: A Dynamical Theory of Structure

The eight results of Sections 3–9 constitute, taken together, a *dynamical theory of structure*: a characterisation of how the structural state $\mathcal{S}(S) = (\bar{\rho}(S), \mathcal{R}(S))$ evolves under fundamental constant deformations. We state the synthesis formally.

Theorem 10.1 (Structural Response Theorem — corpus-supported). *Let S be a physical system with structural state $\mathcal{S}(S)$. The present corpus supports the following five structural response properties under constant deformation operators $\hat{O}_\alpha$ and $\hat{O}_\mu$:*

1. (Operator separation, corpus-level) *On the tested corpus, the μ -operator is realizability-inert while the α -operator generates domain-specific realizability responses; the two operators act on distinguishable structural components.*
2. (Response classification) *Systems can be assigned to realizability response classes $\mathcal{I}_c$, $\mathcal{T}_c$, $\mathcal{F}_c$ for each operator c , based on their behaviour across Γ_d . The classification appears to correlate with the structural organisation of the gap distribution, independently of $\bar{\rho}$, though this connection is empirical.*
3. (Phase-like structure) *The corpus contains evidence of class-change regions in parameter space for selected systems; the precise nature of these changes (sharp transitions vs. continuous crossovers) is not yet determined.*
4. (Metric depth) *Within the FULL realizability class, the connectivity threshold κ_{conn} spans a factor of 5×10^5 at physical parameter values, providing finer structural resolution than the class label alone.*
5. (Corpus-level exclusion) *No system with $\bar{\rho} \geq 0.8$ returns $\mathcal{C} = \text{HARD}$ at $\gamma = 1.00$ in the present corpus.*

Remark 10.2 (Position in the structural hierarchy). The dynamical theory does not replace the static dual-layer theory. It extends it by adding a time-like dimension: instead of characterising $\mathcal{S}(S)$ at a fixed point, it characterises the trajectory $\gamma \mapsto \mathcal{S}(S(\gamma))$ through the structural state space. The static results (USL, PRP, Dual Observability) describe the geometry of that state space; the dynamical theory describes how physical systems move through it under operator deformation.

10.1 The Structural Response Map

Definition 10.3 (Structural response map). For a physical system S and constant c , the *structural response map* is the function

$$\Phi_{S,c} : \Gamma_d \longrightarrow [\bar{\rho}_{\min}, 1] \times \{\text{FULL, GIANT, TAIL, HARD}\} \times (0, \infty],$$

defined by $\Phi_{S,c}(\gamma) = (\bar{\rho}(S(\gamma)), \mathcal{C}(S(\gamma)), \kappa_{\text{conn}}(S(\gamma)))$. The structural response map encodes both the admissibility trajectory (previously studied via STRUC-I) and the realizability trajectory (new, studied here via STRUC-PERC-I).

Proposition 10.4 (Independence of the two trajectories). *The admissibility trajectory $\gamma \mapsto \bar{\rho}(S(\gamma))$ and the realizability trajectory $\gamma \mapsto \mathcal{C}(S(\gamma))$ are not determined by each*

other. A system can have a flat admissibility trajectory (Type I in the USL framework) and a HARD realizability class (hydrogen fs levels), or a curved admissibility trajectory (Type III) and FULL realizability class stable throughout (nuclear isotopes under α).

Proof sketch. Hydrogen fine-structure levels are HARD throughout the α -sweep with constant GR = 0.842: flat admissibility response in the structural sense (the HARD class does not change), but permanent fragmentation. In contrast, ^{48}Ca returns FULL at all tested α -values ($\kappa_{\text{conn}} = 13,240$ at physical) despite being classified Type III-Fr in the USL framework (elevated $\bar{\rho}$, non-monotone under α). The two trajectories are therefore empirically independent. $\square$

11 Discussion and Open Problems

11.1 Implications for the Structural Framework

The dynamical theory adds three new objects to the UNNS structural framework beyond those of the static theory:

1. *The operator algebra* $\{\hat{O}_\alpha, \hat{O}_\mu\}$ acting on the structural state space. The property of operator separation (Theorem 3.1) and the conjecture of non-commutativity (Conjecture 3.4) define the algebraic structure of this set of operators.
2. *The realizability invariance classification*: a partition of physical systems into $\mathcal{I}, \mathcal{T}, \mathcal{F}$ classes for each operator. This classification is orthogonal to the USL TYPE system (which classifies by admissibility trajectory).
3. *The structural response map* $\Phi_{S,c}$: the trajectory of a system through the structural state space under operator deformation. This replaces the one-dimensional admissibility trajectory with a full two-dimensional (admissibility $\times$ realizability) path.

11.2 Relation to the Constant-Anchoring Hypothesis

The constant-anchoring hypothesis of [1] asserts that in domains where a constant is structurally active, the physical value $\gamma = 1.00$ is a structural fixed point of the admissibility geometry. The dynamical theory generates an analogous realizability-theoretic conjecture:

Conjecture 11.1 (Realizability anchoring). In every domain where $\hat{O}_\alpha$ generates a realizability phase transition, the physical value $\gamma = 1.00$ lies in the FULL phase (not in the HARD or TAIL phase). Equivalently: if a system has a deformation-induced HARD window, that window does not contain $\gamma = 1.00$.

Remark 11.2. Conjecture 11.1 is consistent with the corpus: every system in $\mathcal{T}_\alpha$ has its HARD window at non-physical γ -values (^{28}Si at $\gamma \leq 0.90$; ^{238}U at $\gamma = 1.02$). It is also consistent with the Structural Exclusion Principle (Theorem 8.1). However, the corpus does not include deformed-geoid systems at $\gamma = 1.00$, leaving open whether geoid harmonics are FULL at their physical value.

11.3 Open Problems

Problem 1 (Operator composition laws). Measure the joint (α, μ) deformation — sweeping both constants simultaneously — to test Conjecture 3.4 directly. Does $\hat{O}_\alpha \circ \hat{O}_\mu$ produce the same $\mathcal{C}$ trajectory as $\hat{O}_\mu \circ \hat{O}_\alpha$?

Problem 2 (Continuous phase diagrams). The current corpus uses a discrete γ -grid. A fine-grained sweep (e.g. $\delta\gamma = 0.001$) near the phase boundaries of ^{28}Si and ^{238}U would establish whether transitions are sharp (first-order) or gradual (continuous).

Problem 3 (Canonical ladder construction). Definition 7.3 identifies canonical ladders as those minimising outlier counts. A formal criterion for canonical construction, and a proof that the realizability class is canonical-ladder-invariant for a given physical system, would resolve the representation ambiguity demonstrated by helium.

Problem 4 (Geoid harmonics at physical α). The geoid corpus in this paper covers only $\gamma \in \{1.10, 1.20\}$. Testing Earth, Mars, and Moon geoid harmonics at $\gamma = 1.00$ and across the full range $\gamma \in [0.80, 1.20]$ would establish whether geoid systems are in $\mathcal{I}_\alpha$ or $\mathcal{T}_\alpha$, and test Conjecture 11.1.

Problem 5 (α_s and α_G columns). The present corpus covers only α and μ . Extending STRUC-PERC-I to the α_s (strong coupling) and α_G (gravitational coupling) columns would establish whether these metrically-neutral operators (in the admissibility sense) are also realizability-inert — an analogue of the μ -inertness result for the gravitational sector.

Problem 6 (Algebraic closure of operator space). Is the set of structural deformation operators $\{\hat{O}_\alpha, \hat{O}_\mu, \hat{O}_{\alpha_s}, \hat{O}_{\alpha_G}\}$ closed under composition? Does it form a group or monoid with respect to operator composition? These questions open a formal algebraic structure for the constant-operator space of the UNNS framework.

12 Conclusion

The present corpus supports the formulation of a dynamical extension of the dual-layer structural framework [2, 3], in which physical systems are characterised not only by their static structural state but by their trajectories through that state under constant deformation.

What the corpus establishes.

Within the tested μ -deformation corpus (127 runs, molecular and nuclear systems), the μ -operator is realizability-inert: zero HARD verdicts, no class changes at any tested γ -value. Within the tested α -deformation corpus (776 runs), the α -operator produces domain-specific realizability responses: permanent HARD fragmentation in hydrogen fine-structure levels (invariant across the full sweep), representation-sensitive HARD in helium (two of three constructions), class-change regions in ^{28}Si and ^{238}U at non-physical γ , and a HARD regime in geoid harmonics at amplified α .

These observations support a response-class taxonomy $(\mathcal{I}, \mathcal{T}, \mathcal{F})$ that is orthogonal to the admissibility TYPE classification. The FULL realizability class carries internal metric depth: a factor of 5×10^5 in κ_{conn} within a single class. A corpus-level exclusion of the

joint state ($\bar{\rho} \geq 0.8$, HARD) at physical constants is observed; this is consistent with the Forbidden State result of [3] and motivates the realizability anchoring conjecture.

What this paper adds to the framework.

Prior work asked: where is a system in the structural state space? This paper asks: how does it move? The structural response map $\Phi_{S,c}$ provides a two-dimensional trajectory through $(\bar{\rho}, \mathcal{R})$ -space under operator deformation, a new object absent from the static theory. The operator separation result provides the first empirical evidence that the two structural coordinates respond to different constant operators independently.

The operator algebra $\{\hat{O}_\alpha, \hat{O}_\mu\}$ is established at the level of marginal sweeps: inequivalent realizability projections, corpus-level μ -inertness, and conjectured non-commutativity under joint deformation. Whether this algebra extends to a closed algebraic structure remains open.

The consolidated claim.

The present corpus supports the existence of structured dynamical response of physical systems under constant deformation. Within the tested domains: the μ -operator leaves realizability unchanged; the α -operator induces system-dependent realizability evolution; systems partition into response classes; and realizability exhibits both phase-like class-change behaviour and internal metric structure. In all transitional systems, the physical constant value lies in the FULL realizability region — consistent with the realizability anchoring conjecture (Conjecture 11.1), though not yet establishing it.

These observations motivate a dynamical extension of the structural framework, in which systems are characterised by their trajectories under operator deformation as well as by their static structural state.

References

- [1] UNNS Substrate Research Program. *Foundations of the UNNS Substrate: From Universal Admissibility to Structural Regime Theory*. Foundation Document, Intermediate Stage, 2026.
- [2] UNNS Substrate Research Program. *The Percolative Realizability Principle: Realizability Structure of Admissible Configurations and the Revised Necessary Direction*. Working Manuscript (Revised), 2026.
- [3] UNNS Substrate Research Program. *Structural Realizability and Dual Observability in the Admissibility Manifold*. Working Manuscript, 2026.
- [4] UNNS Substrate Research Program. *Cross-Instrument Corpus Analysis: STRUC-I v1.0.4 $\times$ STRUC-PERC-I v2.4.0*. Internal Report, 2026-04-06.
- [5] UNNS Substrate Research Program. *Constants Percolation Corpus Analysis: Realizability Sweep Across α and μ Deformations, 903 Runs*. Internal Report, 2026-04-06.

Table 1: μ -column corpus at physical value ($\gamma = 1.00$). All systems: FULL percolation, zero HARD. κ_{conn} : physical-value connectivity threshold.

System	Domain	Runs	Class at $\gamma = 1.00$	κ_{conn}
H ₂	Molecular	17	FULL	1,422
CO	Molecular	17	FULL	714,804
HCl	Molecular	11	FULL	2
HD	Molecular	17	FULL	10,203
N ₂	Molecular	14	FULL	2
⁴⁸ Ca	Nuclear	17	FULL	13,240
¹⁵⁰ Nd	Nuclear	17	FULL	301
²⁰⁸ Pb	Nuclear	17	FULL	38,013
Total		127	127 FULL / 0 HARD	

A Corpus Summary Tables

A.1 μ -Column Realizability Results (127 runs, all FULL)

A.2 α -Column: Systems with HARD Verdicts

Table 2: α -column systems exhibiting HARD verdicts. HARD at $\gamma = 1.00$: H (fs levels) only. All others: HARD confined to non-physical γ .

System	Domain	Runs	FULL	HARD	HARD at $\gamma = 1.00$?	Notes
H (fs levels)	Atomic	41	0	41	Yes	All γ ; GR = 0.842 constant
He	Atomic	52	18	34	Yes (2 repr.)	Representation-sensitive
²⁸ Si	Nuclear	34	30	4	No	HARD at $\gamma \leq 0.90$, $\gamma = 0.97$
²³⁸ U	Nuclear	33	32	1	No	HARD at $\gamma = 1.02$ only
Geoid Earth	Geoid	2	0	2	n/t	Tested only at $\gamma = 1.10, 1.20$
Geoid Moon	Geoid	2	0	2	n/t	Tested only at $\gamma = 1.10, 1.20$

A.3 κ_{conn} at Physical Value — All FULL Systems

B Acknowledgements

This work is a product of the UNNS Substrate Research Program. All corpus runs were conducted under preregistered falsification protocols. Tier A and proxy-grade results are tracked separately; all deformation rules applied in the α and μ sweeps are documented in the companion Foundations document [1].

Table 3: Connectivity threshold κ_{conn} at $\gamma = 1.00$ for all FULL systems. Sorted ascending. Demonstrates the factor- 5×10^5 internal spread within the FULL class.

System	Domain	Constant	κ_{conn}
HCl	Molecular	μ	2
N ₂	Molecular	μ	2
¹⁵⁰ Nd	Nuclear	α/μ	301
¹⁶⁶ Er	Nuclear	α	440
¹²⁰ Sn	Nuclear	α	1,196
²³⁸ U	Nuclear	α	1,490
H ₂	Molecular	μ	1,422
¹⁷⁴ Yb	Nuclear	α	2,102
Na	Atomic	α	2,446
²⁴ Mg	Nuclear	α	2,771
⁵⁶ Fe	Nuclear	α	3,263
⁹⁰ Zr	Nuclear	α	3,900
²⁸ Si	Nuclear	α	5,299
⁶⁰ Ni	Nuclear	α	6,767
HD	Molecular	μ	10,203
⁴⁸ Ca	Nuclear	α/μ	13,240
¹⁰⁰ Mo	Nuclear	α	35,405
¹⁵² Sm	Nuclear	α	37,087
¹¹⁶ Sn	Nuclear	α	37,217
²⁰⁸ Pb	Nuclear	α/μ	38,013
CMB TE	CMB	α	65,708
Li	Atomic	α	82,359
CMB TT	CMB	α	134,683
CO	Molecular	μ	714,804
CMB EE	CMB	α	859,341
He (repr. 1)	Atomic	α	1,000,000